VERSION 1 · 10 NOVEMBER 2026

How the X-ray *thinks*, and is it ready?

How the logic works step by step, where the data comes from and how it improves, what's ready for launch and what isn't, and an honest look at the revenue.

SHORT ANSWER 1

Nothing is "trained". The X-ray runs on written rules, a dish dictionary and real price data. Every result can be traced back to a rule.

SHORT ANSWER 2

It's ready for a soft launch: real owners, free, Dubai first. It isn't ready for paid scale yet: five gaps remain (page 08).

SHORT ANSWER 3

A million dollars is possible only as a multi-city data product, not as a free tool plus one consultant. The maths is on page 09.

01 · NO MODEL, NO TRAINING

It doesn't learn on its own. *It gets taught.*

An AI model learns patterns from millions of examples and can't explain itself. The X-ray works the other way: a menu engineer's rules, written as code. That makes it predictable, testable and honest about what it doesn't know.

1 · Rules

Around 30 written checks across five parts of a menu: item counts, price ladders, anchors, descriptions, add-ons, local prices, inflation. Each one produces a sentence of evidence.

engine/core/score.ts · xray.ts

2 · Dictionary

133 dishes with their alternative names ("murgh makhani" = butter chicken), usual section, cuisine and a suggested description from the classic recipe.

engine/core/dishes.ts · match.ts

3 · Data

Real public menu prices (560 from 14 Dubai menus) and official restaurant inflation (US BLS, UK ONS). The rules compare against these.

Supabase: benchmark_items · price_index

How it gets better: four feedback loops, all run by people

LOOP	WHAT HAPPENS	WHO / HOW OFTEN
More menus	Public menus are added to the price index; more dishes pass the 5-venue minimum and become comparable.	PIHOCO, ~10 menus a month
Owner corrections	Every saved audit keeps the reviewed menu. Reading where owners fixed rows shows which menu formats the reader gets wrong.	Review monthly; improve the reader
New dishes	Dishes that appear often but aren't in the dictionary (e.g. "V60", "affogato") are added with their aliases.	Monthly, from benchmark rows marked "x:"
Rule tuning	Thresholds (e.g. a café's ideal 18–50 items) are adjusted from what real audits and Piyush's reviews show.	Quarterly; every change gets a test

> Why this matters for marketing: "Rules, not AI" is a trust message. Owners can see exactly why they got their score, and their menus and numbers never train anyone's model.

02 · FROM MENU TO MONEY, IN SEVEN STEPS

Every step is *plain code you can read.*

#	STEP	WHAT THE CODE DOES	OUTPUT
1	**Get text**	Pasted text as is · PDF: pdf.js reads the text layer and rebuilds each line from the position of its words · Photo: Tesseract.js reads the image on the device	Lines of text
2	**Parse**	A price at the end of a line = a dish (handles AED, £, $, ₹, S$, "12 / 15" sizes, "1,250/-", "8,50"). A short capitalised line or a known word = a section. Other lines = descriptions. A price alone = the dish above.	Rows: section, dish, description, price(s)
3	**Flag**	Each row gets ok / check / fix: no price, price far from the rest (×12), very long name, no section above it, name starting with a number	Confidence + reason
4	**Owner reviews**	The owner edits names, prices and sections. Every edit is re-checked.	Clean menu
5	**Recognise**	Each dish is matched to the dictionary (all words of the dish name present; extra words that make it a roll, cake, soda, iced or Spanish version block the match). Protein kept separate.	Dish key, e.g. biryani:lamb
6	**Score**	Five parts scored 0–100 with evidence (page 04). Parts without data are "not assessed"; the total is re-weighted.	Score, coverage, verdict
7	**Find & price**	Fixes are generated from failed checks; money levers use the owner's covers and bill (page 05). Money fixes first, then by priority. Three descriptions rewritten.	Findings, money range, rewrites

Price comparisons

Local prices for the owner's city are loaded before step 6. A dish is compared only if **5+ venues** list it, one price per venue. More than 8% under the median = under-priced, with a test price one 8% step closer.

Inflation drift

If the owner gives their last price change: drift = index now ÷ index then − 1, from the official series for their country. Over 2% becomes a finding.

03 · THE EXACT THRESHOLDS

How each part *scores.*

PART · WEIGHT	STARTS AT · DEDUCTIONS OR POINTS
Structure & balance · 20%	Starts at 100. Too many items for the concept −8 to −35 (restaurant 25–60, café 18–50, bar 15–45, cloud kitchen 15–40, QSR 10–35); too few −10; too many or too few sections −12; each crowded section −10 (max −30); each thin section −5 (max −15)
Price architecture · 25%	Needs 5+ priced dishes. Starts at 100, per section with 3+ prices: flat range (top ÷ bottom < 1.25) −9; very wide (> 3.2×) −7; no anchor (top < 1.3 × median) −6; listed cheapest-first −5. Mixed price endings −8. Currency signs on over half the prices −7.
Descriptions · 15%	Points: share of dishes described (target 70%) up to 40; length 6–22 words 15; taste or method words (target 60%) up to 20; 1–3 chef's picks 15; dietary marks 10
Upsell paths · 20%	Points: add-ons 25; 2+ sides 15; sizes 15; combos or sharing 20; 2+ desserts 10; 4+ drinks 15
Market position · 20%	Local prices (3+ comparable dishes): 100 − average gap × 250. Inflation drift: under 2% = 100, under 5% = 75, under 10% = 50, else 25. Average of what's available; none = not assessed.

Verdicts

- 80+ "A well-built menu. The gains are in the details."
- 65–79 "Solid, with money left on the table."
- 50–64 "Busy, but not earning in the right places."
- Under 50 "This menu is working against you."
- Under 75% coverage adds "Partial read".

Where the numbers come from

Weights come from the PIHOCO product blueprint. Thresholds are menu-engineering practice set as PIHOCO defaults, in one file, so Piyush can tune them. The currency-sign rule cites the Cornell study (Yang, Kimes & Sessarego, 2009).

They are a diagnostic framework, not an industry rating, and every result says so.

04 · HOW MONEY IS ESTIMATED

The owner's numbers, *our visible assumptions.*

Money appears only if the owner gives covers per week and average bill. Monthly revenue = covers × bill × 52 ÷ 12. Every lever is a **range**, every assumption is a slider, and it's always labelled "Scenario, not a forecast".

LEVER	FORMULA, PER MONTH	DEFAULT ASSUMPTIONS
Add-on line	covers × take-up × (bill × 12%) × 4.33	Take-up 4–10% · shown when there are no add-ons
Dessert / drink close	covers × take-up × (bill × 18%) × 4.33	Take-up 3.2–8% · shown when desserts < 2 or drinks < 4
Inflation catch-up	revenue × min(drift, 6%) × 75% kept; low end uses half the step	Cap 6% per step · 25% of the gain lost to fewer orders
Under-priced dishes	(test price − price) × weekly units × 4.33; low end 60%	Only with weekly units from the owner
Total	Sum of levers; high end × 0.85 for overlap	Rounded to 10 / 50 / 500 steps

Worked example (built-in sample menu)

```
Revenue: 600 covers × AED 85 × 52 ÷ 12 = AED 221,000 a month
Add-on: 600 × 4% × 10.20 × 4.33 = 1,061 → 1,050 · 600 × 10% × 10.20 × 4.33 = 2,652 → 2,650
Dessert close: 600 × 3.2% × 15.30 × 4.33 = 1,273 → 1,250 · 600 × 8% × 15.30 × 4.33 = 3,182 → 3,200
Total: 1,250 + 1,050 = 2,300 · (3,200 + 2,650) × 0.85 = 4,973 → 4,950 → "+AED 2,300–4,950 a month"
```

> What it never does: call any of this profit, invent sales volumes, or claim results. Profit needs recipe costs and sales mix, which is the paid audit's job.

05 · THE DATA AND ITS MONTHLY CYCLE

The data is the moat, *so it has a routine.*

560	14	22	42
DUBAI MENU PRICES	PUBLIC MENUS	COMPARABLE DISHES	INFLATION DATA POINTS

DATASET	BUILT HOW	QUALITY RULES
Dubai price index	Copied from 14 venues' own websites on 10 Nov 2026 (4 restaurants, 10 cafés); dish keys assigned by the same matcher the app uses	Source link per price · no delivery apps · one price per venue per dish · 5-venue minimum · grouping checked by hand
Owner contributions	Price Check, when the owner ticks "add to the index"	Stored unverified · reviewed before counting in claims
US inflation	BLS "food away from home": FRED levels + BLS monthly and 12-month changes	Mar 2025 – Sep 2026
UK inflation	ONS division 11 annual rates, chained monthly	Labelled an estimate · Jan 2024 – Sep 2026
Dish dictionary	Written by PIHOCO from classic recipes	133 dishes · 13 automated tests, including a 17-case matching test

The monthly routine (about 3 hours)

01 Add ~10 public menus to `data/`, run the seed script; check the dish groups it prints.

02 Add the new BLS and ONS month to `price_index`.

03 Review owner-contributed prices; mark the good ones verified.

04 Scan saved audits for misread menus and unknown dishes; add to the dictionary.

05 Republish the price index posts with the new month and counts.

06 · WHAT'S BEEN CHECKED, AND WHAT IT CAN'T DO

Honest about *its edges.*

Checked

- 13 automated engine tests pass: parsing formats, money maths by hand, re-weighting, 5-venue minimum, dish matching including false matches
- Strict TypeScript build passes
- Full flow tested in a browser at phone and desktop sizes, against the real Dubai data
- Database rules tested as a public user: leads and audits unreadable, no-consent leads refused
- An audit of every dish grouping in the data found and fixed false matches (e.g. "Tea" → steak) before launch

Limits today

- **Photos**: decorative fonts read badly. The review step catches it, but it's friction.
- **Restaurants**: no restaurant dish has 5 venues yet, so market comparison works for cafés only.
- **Countries**: inflation for US and UK only; UAE, India and Singapore not yet.
- **Languages**: English menus. Arabic and Hindi names work only where the dictionary has them in Latin letters.
- **No profit view**: no recipe costs or sales mix; by design, that's the audit.
- **Untested live**: no real owner has used the live app yet.

> Rule for the team: when a result looks wrong, find the rule or the data row that caused it, fix it, and add a test with that menu. That's the whole "training" process.

07 · IS IT READY TO LAUNCH?

Ready for a *soft launch.* Not yet for paid scale.

ITEM	STATUS	WHAT'S NEEDED
App live, mobile-first, on-brand	**Ready**	pihoco.vercel.app/app
Engine, tests, dish matching	**Ready**	13 tests passing
Database and security rules	**Ready**	Own Supabase, tested as public user
Website section and nav links	**Ready**	Homepage and cafés page
Content bank, pricing, deck, privacy doc	**Ready**	Delivered
Live lead test (save + email)	**To do · 5 min**	One test from a phone; confirm the email arrives
Brand domain	**To do · 15 min**	pihoco.com pointed at Vercel
Database always on	**Before promotion**	Supabase Pro (~USD 25/mo): free projects pause after a week idle
Privacy page covers the X-ray	**Before promotion**	Use pages 02–05 of the data PDF
Restaurant comparisons	**Gap**	~25 more restaurant menus (Indian, Lebanese first)
Payments for paid tiers	**Before 1 Feb**	Stripe or Ziina payment link for the USD 49 scorecard
Analytics	**Recommended**	Privacy-friendly page analytics (e.g. Plausible) to measure the funnel
Email follow-ups	**Manual for now**	Day 2–30 emails sent by hand until volume justifies automation

Soft launch = real owners, free, Dubai cafés first, Piyush reviewing every scorecard. Use it for 4–8 weeks to learn which findings owners act on before charging.

08 · CAN IT MAKE USD 1,000,000?

Not as it stands. *Here's the maths.*

USD 1M a year means about USD 83,000 every month. On the current pricing, any one product alone would need:

REVENUE LINE	PRICE	NEEDED FOR USD 1M / YEAR	REALITY CHECK
Menu Profit Boost Audit	USD 750–1,200	~1,000 audits (~85 a month)	One consultant delivers ~4–8 a month. Capped at ~USD 50–110k a year.
Full Scorecard	USD 49	~20,400 a year	Each is reviewed by hand: needs a team or automation.
Price Watch	USD 19 / month	~4,400 paying venues	Needs deep data in many cities and low churn.
Group / consultant plan	USD 99 / month	~840 accounts	Needs multi-location features not built yet.

Path A · Consulting engine

The X-ray feeds Piyush's audits and sprints.

Realistic year one: USD 50–150k. Low cost, fast, proven model.

Path B · Data product

The price index in 5–10 GCC and UK cities, sold as Price Watch and reports to groups, suppliers and landlords.

USD 1M is possible in about 3 years with ~1,500–3,000 menus per city and a small team.

Path C · Licensing

White-label the X-ray for POS companies, delivery platforms or F&B consultancies.

A few deals at USD 20–100k each, once the data and track record exist.

> Recommendation: run Path A now to fund and validate; build the data for Path B in parallel. Don't spend on scale until 50+ real X-rays show owners act on the findings.

09 · FROM LAUNCH TO PROOF

Three months, *five numbers.*

WHEN	DO	SUCCESS LOOKS LIKE
Week 1	Live lead test · domain · Supabase Pro · privacy page · analytics	Every item on page 08 that says "before promotion" is done
Weeks 2–4	Soft launch to Dubai cafés: price index posts, 20 DMs a week, Piyush reviews every scorecard	50 X-rays · 10 scorecard requests
Weeks 5–8	+25 restaurant menus; first anonymised teardown; ask owners which fix they made	Restaurant comparisons live · 2 audits sold
Weeks 9–12	Turn on the USD 49 scorecard (from 1 Feb); test a Price Watch waitlist	150 X-rays · 5 audits · 30 on the Price Watch waitlist

The five numbers to watch every week

X-rays
LIVE COUNTER IN THE DATABASE

15–25%
RESULT → SCORECARD REQUEST

20%
SCORECARD → AUDIT

+10
MENUS ADDED PER MONTH

Fifth number: the share of owners who say they **changed something** after the X-ray. That is the proof every later sale, and any million-dollar plan, depends on.

> If after 90 days owners run it but don't act, the product needs to change before it scales. If they act and buy audits, scale the data (Path B) with confidence.